

Checkpoint B Research Report

QuantaFlex ETF Data Preparation

Caleb Boampong

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Abstract

QuantaFlex ETF is a proposed rules-based, actively managed multi-asset ETF investing across equities, bonds, commodities, and cash. This checkpoint establishes the historical data, asset proxies, return calculations, and evaluation framework needed to test the strategy. Daily adjusted price data were collected from Yahoo Finance using the Python package `yfinance`, and daily log returns were calculated for analysis. Because the selected ETF proxies do not all exist back to 1999, the main common sample is 2008–2024. This period still captures major market disruptions, including the global financial crisis and COVID-19. Initial results from a long-only buy-and-hold portfolio and a basic Monte Carlo simulation show that the project is feasible and ready for expanded backtesting.

Keywords: ETF, Monte Carlo, backtesting, log returns, momentum

Introduction

This research is being conducted to evaluate whether QuantaFlex ETF can be implemented as a transparent, rules-based, multi-asset investment fund with practical risk controls. Many investors want diversified active management, but traditional active funds are often expensive, opaque, and heavily dependent on manager discretion. QuantaFlex is intended to offer a more systematic alternative.

The intended users are retail investors, financial advisors, and institutional investors seeking liquid, diversified, and rules-based exposure. The application being developed is a portfolio research framework that can collect market data, compute returns and indicators, test allocation rules, and evaluate investment performance.

The main research question for this checkpoint is: What securities, data sources, and time-series design choices are appropriate for evaluating QuantaFlex ETF?

Literature Review

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The project is supported by prior research in portfolio theory, asset pricing, momentum investing, and backtesting. Markowitz (1952) provides the basis for diversification and portfolio construction. Sharpe (1964) supports the use of benchmark-relative performance measures such as alpha and beta. Momentum and trend-following studies suggest that past price behavior can be useful in forming systematic trading strategies across multiple asset classes (Jegadeesh and Titman 1993; Moskowitz, Ooi, and Pedersen 2012). Goldfarb and Iyengar (2003) further show that portfolio selection should account for uncertainty in the inputs.

For model evaluation, López de Prado (2018) argues that historical backtesting alone can be misleading if the strategy is overfit to one market path. Monte Carlo simulation provides a way to test strategies across many possible scenarios, which makes it useful for evaluating a proposed ETF that does not yet have a live performance record.

Methods

The purpose of this checkpoint was to identify the historical data required to evaluate QuantaFlex ETF and to build an initial baseline portfolio.

The selected securities were NVDA for U.S. growth equity exposure, AGG for U.S. investment-grade bonds, ACWI for global equities, GLD for gold, DBC for broad commodities, BIL for cash or Treasury-bill reserves, and SPY as the market benchmark. These securities were selected because they are liquid, publicly traded, and represent the major asset classes in the QuantaFlex concept.

The main data source was Yahoo Finance through the Python package `yfinance`. Daily adjusted close prices were downloaded for the period 2008–2024. Adjusted prices were used because they incorporate dividends and stock splits, making them more appropriate for total-return analysis.

Daily log returns were calculated using the standard formula:

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$$r_t = 100 \times [\ln(P_t) - \ln(P_{t-1})]$$

where P_t is the adjusted closing price on day t .

Following the professor's recommendation to begin simply, the initial strategy was a long-only, equal-weight buy-and-hold portfolio across NVDA, AGG, ACWI, GLD, DBC, and BIL. SPY was used as the comparison benchmark. Performance was evaluated using annualized return, annualized volatility, Sharpe ratio, maximum drawdown, alpha, and beta. A basic Monte Carlo simulation using a multivariate normal return model was also run over a five-year horizon with 500 simulated paths.

Results

The data download and preprocessing were successful for all selected tickers. A usable common dataset was constructed for 2008–2024, and daily log returns were computed for each security. Summary statistics showed meaningful differences across the assets, with NVDA having the highest average daily return and much higher volatility than the bond and cash proxies. Defensive assets such as AGG and BIL showed much lower volatility, while GLD and DBC provided commodity-related diversification.

The normalized price chart showed that NVDA strongly outperformed the other assets over the sample period, while the remaining ETF proxies had more moderate growth paths. This confirms that the portfolio contains both high-growth and defensive components.

The equal-weight QuantaFlex buy-and-hold portfolio produced the following performance results:

- Annualized Return: approximately 8.03%
- Annualized Volatility: approximately 12.92%
- Sharpe Ratio: approximately 0.62
- Maximum Drawdown: approximately -39.54%

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- Alpha: approximately 2.64%
- Beta: approximately 0.50

The SPY benchmark produced:

- Annualized Return: approximately 10.85%
- Annualized Volatility: approximately 19.95%
- Sharpe Ratio: approximately 0.54
- Maximum Drawdown: approximately -51.48%

These results suggest that the QuantaFlex buy-and-hold portfolio underperformed SPY in raw return, but it also had substantially lower volatility, lower drawdown, and a slightly higher Sharpe ratio. Its beta of about 0.50 indicates that it moved with the market, but with much lower overall exposure. This supports the idea that the multi-asset structure provides diversification benefits.

The Monte Carlo simulation produced a mean five-year ending portfolio value of about 1.55 and a median ending value of about 1.50, starting from an initial value of 1.00. This suggests positive average growth under simulated conditions, although the distribution also showed downside and upside variation. The simulation confirms that the portfolio framework can be tested under uncertainty, which is important for the later evaluation of active rules.

Conclusions

Checkpoint B successfully established the data foundation for QuantaFlex ETF and produced a working baseline portfolio. The selected asset universe, public data source, return methodology, and benchmark comparison are all in place. The initial results show that the diversified buy-and-hold portfolio has lower risk and shallower drawdowns than SPY, although at the cost of lower total return. This is consistent with the goal of building a multi-asset ETF with embedded risk control.

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The main limitation is that the chosen ETF proxies do not support a full 1999–2024 common history, so the analysis uses 2008–2024 instead. Another limitation is that the current model is only a passive equal-weight baseline and does not yet include the active QuantaFlex rules such as moving-average signals, momentum tilts, or volatility-based exposure adjustments.

Overall, the project appears feasible and is progressing appropriately. The next steps are to implement the active allocation rules, introduce walk-forward backtesting, test parameter sensitivity, and refine the Monte Carlo approach so that it better preserves time-series dependence. These steps will move the project from data preparation to full strategy testing.

References

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